

Michael Zingale / Curriculum Vitæ

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Present Position

Sept. 2021– *Professor of Physics and Astronomy, Stony Brook University, Stony Brook, NY*

Research Interests

I am interested in developing and applying computational hydrodynamics algorithms to problems in nuclear astrophysics. A large part of this work is the development of low Mach number hydrodynamics code MAESTROeX and the compressible (magneto-, radiation-) hydrodynamics code Castro. Both codes are freely available on github, use adaptive mesh refinement, and hybrid parallelism techniques to run at scale on today's supercomputers. I apply these codes to studies of X-ray bursts, different progenitor models of Type Ia supernovae, and convection in stars. Importantly, all of the code, input files, workflow scripts needed to reproduce the science done in my research group is available in our github repos.

Education

- 2000 Ph.D. in Astronomy and Astrophysics, University of Chicago
thesis: Helium Detonations on Neutron Stars
advisor: Dr. J. W. Truran
- 1998 M.S. in Astronomy and Astrophysics, University of Chicago
- 1996 B.S. in Physics and Astronomy, University of Rochester, Magna Cum Laude
thesis: Magnetohydrodynamical Wave Support of Molecular Clouds
Minor in Mathematics, University of Rochester

Academic Appointments

- 2021– *Astronomy Undergraduate Advisor, Stony Brook University*
- 2014– *Affiliate, Institute for Advanced Computational Science, Stony Brook University*
- 2012–2021 *Associate Professor of Physics and Astronomy, Stony Brook University*
- 2006–2011 *Assistant Professor of Physics and Astronomy, Stony Brook University*
- 2001–2005 *Postdoctoral Researcher, SciDAC Supernova Science Center, University of California, Santa Cruz. Worked on simulations of turbulent thermonuclear flames in Type Ia supernova. Initiated a collaboration with Lawrence Berkeley Lab to apply low Mach number hydrodynamics methods to astrophysical flames. advisor: Dr. S. E. Woosley*
- 2000–2001 *Research Associate, Center for Astrophysical Thermonuclear Flashes, University of Chicago. One of the developers of the FLASH Code. Research focused on flame simulations in Type Ia supernovae. advisor: Dr. J. W. Truran*

1997–2000 *Graduate student researcher, Center for Astrophysical Thermonuclear Flashes and Department of Astronomy and Astrophysics, University of Chicago. One of the developers of the FLASH Code. advisor: Dr. J. W. Truran*

Honors / Awards

2022 *Outstanding Faculty Award, Stony Brook University, Department of Physics and Astronomy*

2019 *Godfrey Excellence in Teaching Award, Stony Brook University, College of Arts and Sciences*

2015–2016 *Scialog Fellow for Scialog: Time Domain Astrophysics: Stars and Explosions*

2006 *Presidential Early Career Award in Science and Engineering (PECASE). Nomination through DOE NNSA.*

2006 DOE Office of Nuclear Physics *Outstanding Junior Investigator (OJI)* Award for a proposal entitled: *Multidimensional Modeling of Astrophysical Thermonuclear Explosions*

2000 *Gordon Bell Award in High Performance Computing, Special Category for a paper entitled High-Performance Reactive Fluid Flow Simulations Using Adaptive Mesh Refinement on Thousands of Processors, Calder et al. 2000. (SC 2000 conference)*

2000 *Carl Sagan Award for Excellence in Teaching (Dept. of Astronomy & Astrophysics, University of Chicago)*

1997 *Gregor Wentzel graduate teaching award (Dept. of Physics, University of Chicago)*

1996 *Stoddard Prize in physics for senior thesis (University of Rochester)*

1996 *Flagg Award for highest GPA in physics (University of Rochester)*

1996 *Inducted into Phi Beta Kappa honor society (University of Rochester)*

1994 *Inducted into Sigma Pi Sigma physics honor society (University of Rochester)*

Publications

100+ refereed publications and conference proceedings

Research Grants/Contracts as Principal Investigator

2025–2028	Department of Energy, Office of Nuclear Physics, <i>Research in Nuclear Astrophysics: Supernovae, Compact Objects, and Algorithms</i> , DOE DE-FG02-87ER40317, Co-Is: Alan Calder, James Lattimer	\$500,000
2024–2026	Research Foundation of SUNY, Seed Funding, <i>Building a Community for Reactive Flow Software Infrastructure</i>	\$50,000
2025	Department of Energy, Office of Nuclear Physics, <i>Research in Nuclear Astrophysics: Supernovae, Compact Objects, and Algorithms</i> , supplement request DOE DE-FG02-87ER40317, Co-Is: Alan Calder, James Lattimer	\$100,000 (additional supplement)

2024–2025	Department of Energy, Office of Nuclear Physics, <i>Research in Nuclear Astrophysics: Supernovae, Compact Objects, and Algorithms</i> , supplement request DOE DE-FG02-87ER40317, Co-Is: Alan Calder, James Lattimer	\$110,000 + \$10,000 (two supplements)
2021–2024	Department of Energy, Office of Nuclear Physics, <i>Research in Nuclear Astrophysics: Supernovae, Compact Objects, and Algorithms</i> , DOE DE-FG02-87ER40317, Co-Is: Alan Calder, James Lattimer	\$1,095,000
2020–2023	Contract with Lawrence Berkeley National Laboratory (part of the DOE ECP Exastar project), contract # 7418390, Co-I: Alan Calder	\$580,951
2018–2019	Contract with Lawrence Berkeley National Laboratory (part of the DOE ECP Exastar project), contract # 7418390, Co-I: Alan Calder	\$144,588
2017–2022	Department of Energy, Office of Nuclear Physics & Office of Advanced Scientific Computing Research, <i>Towards Exascale Astrophysics of Mergers and Supernovae (TEAMS)</i> (SBU subcontract through MSU, multi-institution collaboration, DE-SC0017955), Co-Is: Alan Calder, James Lattimer	\$616,000
2011–2013	Department of Energy, Office of Nuclear Physics (2.5-year renewal), <i>Multidimensional Modeling of Astrophysical Thermonuclear Explosions</i> , DOE DE-FG02-06ER41448	\$253,000
2010–2011	Contract with Lawrence Livermore National Laboratory, <i>Multidimensional Modeling of Nova with Realistic Nuclear Physics</i> , 2010: B589924; 2011: B593287	\$99,768
2009–2011	Department of Energy, Office of Nuclear Physics Outstanding Junior Investigator Award (2-year renewal), <i>Multidimensional Modeling of Astrophysical Thermonuclear Explosions</i> , DOE DE-FG02-06ER41448	\$186,000
2007–2009	Contract with Lawrence Livermore National Laboratory, <i>Verification and Validation of Radiation Hydrodynamics for Astrophysical Applications</i> , 2007: B568673; 2008: B574691; 2009 B582735	\$150,000
2006–2009	Department of Energy, Office of Nuclear Physics Outstanding Junior Investigator Award, <i>Multidimensional Modeling of Astrophysical Thermonuclear Explosions</i> , DOE DE-FG02-06ER41448	\$255,000
Research Grants/Contracts as Co-Investigator		
2019–2022	National Science Foundation, <i>REU Site: Broadening undergraduate research participation in Physics and Astronomy at Stony Brook University</i> , NSF 1852143, PI: Matthew Dawber, Co-Is: Navid Vafael-Najafabadi, Michael Zingale	\$273,308

2018–2021	Department of Energy, Office of Nuclear Physics, <i>Research in Nuclear Astrophysics: Supernovae, Compact Objects, and Algorithms</i> , DOE DE-FG02-87ER40317, PI: James Lattimer, Co-Is: Alan Calder, Michael Zingale	\$1,140,000
2015–2018	Department of Energy, Office of Nuclear Physics, <i>Research in Nuclear Astrophysics: Supernovae, Compact Objects, and Algorithms</i> , DOE DE-FG02-87ER40317, PI: James Lattimer, Co-Is: Alan Calder, Michael Zingale	\$1,100,000
2013–2015	Department of Energy, Office of Nuclear Physics <i>Research in Nuclear Astrophysics: Supernovae, Compact Objects, and Algorithms</i> , DOE DE-FG02-87ER40317, PI: James Lattimer, Co-Is: Alan Calder, Michael Zingale	\$640,000
2012–2015	NSF, <i>White Dwarf Mergers as Progenitors of Type Ia Supernovae</i> , AST-1211563, PI: Alan Calder, Co-Is: Doug Swesty, Michael Zingale	\$437,643

Large Computer Time Allocations

2026	PI on a NERSC 2026 allocation, <i>Three-dimensional studies of white dwarfs, massive stars, and neutron star systems</i> (50k CPU node-hours; 100k GPU node-hours)
2025	PI on a NERSC 2025 allocation, <i>Three-dimensional studies of white dwarfs, massive stars, and neutron star systems</i> (50k CPU node-hours; 50k GPU node-hours)
2024	PI on a NERSC 2024 allocation, <i>Three-dimensional studies of white dwarfs, massive stars, and neutron star systems</i> (100k CPU node-hours; 100k GPU node-hours)
2023	PI on a NERSC 2023 allocation, <i>Three-dimensional studies of white dwarfs, massive stars, and neutron star systems</i> (100k CPU node-hours; 150k GPU node-hours)
2023–	PI on an INCITE 2023 award, <i>Exascale Models of Astrophysical Thermonuclear Explosions</i> (2023: 400 k node hours on OLCF summit, 300 k node hours on OLCF frontier, 100 k node hours on ALCF polaris; 2024: 400 k node hours on OLCF frontier, 75 k node hours on ALCF polaris; 450 k node hours on OLCF frontier, 50 k node hours on ALCF polaris)
2022	PI on a NERSC 2022 allocation, <i>Three-dimensional studies of white dwarfs, massive stars, and neutron star systems</i> (80k CPU node-hours; 100k GPU node-hours)
2021–2022	PI on an INCITE 2021 award <i>Approaching Exascale Models of Astrophysical Explosions</i> (2021: 700 k node hours on OLCF summit; 2022: 590 k node hours on OLCF summit, 100 k node hours on ALCF polaris)
2020	PI on a NERSC 2021 allocation, <i>Three-dimensional studies of white dwarfs, massive stars, and neutron star systems</i> (30 M MPP hours)
2019	PI on a NERSC 2020 allocation, <i>Three-dimensional studies of white dwarfs, massive stars, and neutron star systems</i> (30 M MPP hours)
2019–2020	PI on an INCITE 2019 award for at OLCF, <i>Approaching Exascale Models of Astrophysical Explosions</i> (2019: 1.5 M node hours on titan, 105 k node hours on summit; 2020: 300 k node hours on summit)

2019	PI on a NERSC 2019 allocation, <i>Three-dimensional studies of white dwarfs, massive stars, and neutron star systems</i> (27.5 M MPP hours)
2018	PI on a NERSC 2018 allocation, <i>Three-dimensional studies of white dwarf and neutron star systems</i> (20.85 M MPP hours)
2018	PI on an INCITE 2018 award for at OLCF, <i>Approaching Exascale Models of Astrophysical Explosions</i> (40 M hours)
2017	PI on a NERSC 2017 allocation, <i>Three-dimensional studies of white dwarf and neutron star systems</i> (5 M MPP hours)
2017	PI on an INCITE 2017 award for the OLCF Cray XKT titan machine, <i>Approaching Exascale Models of Astrophysical Explosions</i> (45 M hours)
2016	PI on a NERSC 2016 allocation, <i>Three-dimensional studies of neutron star systems</i> (4.6 M MPP hours)
2015–2016	PI on an INCITE 2015 award for the OLCF Cray XK7 titan machine, <i>Approaching Exascale Models of Astrophysical Explosions</i> (2015: 50 M hours, 2016: 55 M hours)
2011–2015	Co-I on NSF PRAC for NCSA/Blue Waters, <i>Type Ia Supernovae</i> (9.1 M node hours)
2015	PI on a NERSC 2015 allocation, <i>Three-dimensional studies of convection in X-ray bursts</i> (5.9 M MPP hours)
2014	PI on a NERSC 2014 allocation, <i>Three-dimensional studies of convection in X-ray bursts</i> (14 M MPP hours)
2014	Co-I on a NERSC 2014 allocation, <i>Type Ia Supernovae and X-Ray Bursts</i> (9 M MPP hours)
2012–2014	Co-I on an INCITE 2012 award for the OLCF Cray XT5, <i>Petascale Simulations of Type Ia Supernovae</i> (2012: 46 M hours; 2013: 55 M hours; 2014: 50 M hours)
2013	PI on XSEDE allocation on Kraken/NICS, <i>CASTRO Simulations of Merging White Dwarfs</i> (4.1 M hours)
2013	Co-I on a NERSC 2013 allocation, <i>Type Ia Supernovae and X-ray Bursts</i> (3.5 M MPP hours)
2011	Co-I on a TeraGrid allocation on the Kraken machine, <i>Thermonuclear Bursts on the Surfaces of Compact Astrophysical Objects</i> (2.1 M hours, Oct. 2011)
2011	Co-I on an INCITE 2011 award for the Cray XT5/ORNL machine, <i>Petascale Simulations of Type Ia Supernovae</i> (50 M hours)
2010	PI on a TeraGrid allocation on the Kraken machine, <i>Thermonuclear Bursts on the Surfaces of Compact Astrophysical Objects</i> (1 M hours; Oct. 2010)
2010	Co-I on an INCITE 2010 award for the Cray XT5/ORNL, <i>Multidimensional Models of Type Ia Supernovae from Ignition to Observables</i> (5 M hours initially + 20 M hours supplement)
2007–2009	Co-Investigator on an INCITE 2007 award for the Cray XT3/ORNL, <i>First Principles Models of Type Ia Supernovae</i> . (2007: 4 M hours; 2008: 3.5 M hours; 2009: 3 M hours)
2006	Co-Principal Investigator on the Leadership Computing Facility (ORNL) allocation, <i>Ignition and Flame Propagation in Type Ia Supernovae</i> . (3 M hours)

Stony Brook Physics and Astronomy Teaching Experience*Undergraduate*

<i>Astronomy Today</i> (AST 100)	A one-credit undergraduate seminar on current astronomy topics, where students lead the discussion on current topics. (F 2010, F 2011, F 2014, F 2015, F 2020, F 2022, F 2024)
<i>Introduction to the Solar System</i> (AST 105)	An overview of solar system topics (solar system dynamics, Kepler's laws, planetary processes, exoplanets, ...) for non-majors. (F 2007, F 2008, F 2009, F 2011, S 2014, S 2015)
<i>Astronomy</i> (AST 203)	A calculus-based introduction to astronomy and astrophysics for majors, covering the basics of radiation, spectra, binary stars, stellar evolution, ISM, clusters, galaxies, and cosmology. (S 2007, S 2008, S 2009, S 2010, S 2011, S 2012, S 2017, S 2019)
<i>Introduction to Planetary Sciences</i> (AST 205)	A calculus-based introduction to the solar system for majors covering basic solar system motion, planetary processes, exoplanets, and solar system formation. (F 2010, F 2014, F 2016)
<i>Stars and Radiation</i> (AST 341)	An overview on stellar physics for undergraduate astronomy majors. (F 2018, F 2020, F 2022, F 2024)
<i>Special Topics: Computational Astrophysics</i> (AST 390)	An introduction to numerical methods used throughout computational astrophysics. (S 2023, S 2025)
<i>Computation for Physics and Astronomy</i> (PHY 277)	An introduction to the Unix command line and a modern compiled programming language. (S 2026)

Graduate

<i>Computational Methods in Physics and Astrophysics I</i> (PHY 504)	An introduction to the Unix command line, version control, C++, data structures, and scientific programming. (S 2022, S 2024)
<i>Stars</i> (PHY 521)	A graduate-level introduction to the physical processes inside stars, stellar structure and atmospheres, and stellar explosions. (F 2013, F 2015, F 2021, F 2023, F 2025)
<i>Python for Scientific Computing</i> (PHY 546; formerly grad special topics)	A one-hour weekly graduate seminar that I created that introduces python and a variety of libraries (NumPy, matplotlib, SciPy, SymPy) for numerical analysis, visualization, and data processing, as well as basic software engineering practices (git/github, debugging, testing). (S 2014, S 2015, S 2016, S 2017, S 2018, S 2022, S 2023, S 2024, S 2025, S 2026)

<i>Computational Methods in Physics and Astrophysics II</i> (PHY 604; formerly grad special topics)	A practical introduction to good development practices, order-of-accuracy, numerical differentiation, integration, interpolation, ODEs, root finding, fitting, FFTs, Monte Carlo, solving hyperbolic, elliptical, and parabolic PDEs, computational fluid dynamics, and parallel programming, with examples in python. (S 2013, S 2016, F 2017)
<i>Astrophysical Fluids and Plasmas</i> (grad special topics)	An introduction to hydrodynamics, fluid instabilities, applications to astrophysics, and an introduction to MHD. (S 2018, S 2021)
<i>The Application of Simulation in Astrophysics</i> (grad special topics)	Develop the equations of hydrodynamics, instabilities common in astrophysics, and discuss numerical methods for solving the Euler equations (finite-volume methods, Riemann solvers, etc.) (S 2006)

Other Teaching Experience

Summer 2026	Lecturer for the MESA Summer School 2026 (Jackson, WY), <i>All-in on Urca</i> (https://zingale.github.io/mesa-labs)
Summer 2023	Lecturer for the Flatiron/CCA Fluid Dynamics Summer School on <i>Coding Solvers for Fluids</i> (https://zingale.github.io/cca-summer-school)
Fall 2020	Instructor for Software Carpentry training event at Institute for Advanced Computational Science, Stony Brook, NY (taught: bash, git)
Summer 2020–2025	Developed and led the Physics and Astronomy REU <i>Python Tutorial</i> of introductory tutorials and exercises over the 10 week program (https://github.com/sbu-phy-ast-reu/reu-python-tutorial).
Feb 2019	Instructor for Software Carpentry training event at Institute for Advanced Computational Science, Stony Brook, NY (taught: python, git).
Summer 2017	Developed and led the <i>Python Boot Camp</i> week-long tutorial for the IACS Data + Compute = Discovery Research Experience for Undergraduates program (https://sbu-python-summer.github.io/)
Summer 2001	<i>University of Chicago / Department of Computer Science:</i> Teaching assistant for the Introduction to Programming in C class in the Computer Science Professional Masters Program at the University of Chicago.
1997–1998	<i>Center of Astronomical Research in Antarctica (CARA) outreach program:</i> Developed and taught thermodynamics, E&M, and mechanics experiments to grade 7–12 Chicago school students. Awarded the Carl Sagan teaching award.
1996–1997	<i>Introductory Physics Teaching Assistant (University of Chicago):</i> Taught weekly discussion and laboratory sections. Awarded the Gregor Wentzel teaching award.

Professional Development

2018	Software Carpentry instructor certification
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- 2001 student at Finite Volume Upwind and Centered Methods for Hyperbolic Conservation Laws (Barcelona, Spain)
- 1999 student at NASA Summer School for High Performance Computational Earth and Space Sciences

Stony Brook Physics and Astronomy Service

- 2026 Title II ADA Compliance committee
- 2020– Astronomy Director of Undergraduate Studies
- 2020– Undergraduate Curriculum Committee, Dept. of Physics and Astronomy
- 2017– Undergraduate Research Committee, Dept. of Physics and Astronomy
- 2024 Promotion committee for appointment to Professor for Department colleague, Dept. of Physics and Astronomy
- 2021 Promotion committee for appointment to Professor for Astronomy colleague, Dept. of Physics and Astronomy
- 2018–2021 Diversity Committee, Dept. of Physics and Astronomy (chair: 2019, 2021)
- 2019 Three-year Reappointment Committee for Physics colleague, Dept. of Physics and Astronomy (chair)
- 2011–2012, Strategic Advising Committee, Dept. of Physics and Astronomy
2013–2019
- 2017 Tenure Committee for Astronomy colleague, Dept. of Physics and Astronomy
- 2006–2007, Graduate Admission Committee, Dept. of Physics and Astronomy
2016–2017
- 2016–2017 Examine the Graduate Exam Committee, Dept. of Physics and Astronomy
- 2013–2016 Astronomy Open Nights coordinator, Dept. of Physics and Astronomy
- 2008, Department Chair Search Committee, Dept. of Physics and Astronomy
2014–2015
- 2014–2015 Three-year Reappointment Committee for astronomy colleague, Dept. of Physics and Astronomy
- 2013–2014 Undergraduate Astronomy Coordinator, Dept. of Physics and Astronomy
- 2013–2014 Tenure Committee for Astronomy colleague, Dept. of Physics and Astronomy
- 2013–2014 Astronomy Faculty Search Committee, Dept. of Physics and Astronomy
- 2013 Ad-hoc Committee for High-Energy Physics Hire, Dept. Physics and Astronomy
- 2007–2012 Colloquium Committee, Dept. of Physics and Astronomy (chair: Fall 2008, Fall 2009, Fall 2010, Fall 2011)
- 2011 CESAME/Physics and Astronomy joint hire committee, Dept. of Physics and Astronomy
- 2009 Long Range Planning Committee, Dept. of Physics and Astronomy

2007–2009	Graduate Advising Committee, Dept. of Physics and Astronomy
2007–2008	Astronomy Faculty Search Committee, Dept. of Physics and Astronomy
2006–2007	NYCCS Faculty Search Committee (Dept. level), Dept. of Physics and Astronomy

Stony Brook University Service

2026	Member of IACS Faculty Search Committee, Institute for Advanced Computational Science
2025	IACS Jr. Research Award Committee, Institute for Advanced Computational Science
2025	Review Panelist for OSP Internal Competition DOE EXPRESS: 2025 Exploratory Research for Extreme-Scale Science
2010	Teaching Learning Technology (TLT) Advisory Committee
2006–2009	University Senate Committee on Computing and Communications (chair: Feb. 2008 – May 2009)

Professional Service

2025–2026	Member of DOE Office of Nuclear Physics, Nuclear Theory (virtual) comparative review panel.
2025–2026	Member of the APS Hans Bethe Prize Committee
2024–	Member-at-Large for APS Division of Computational Physics Executive Committee
2024	Member of the APS DCOMP Metropolis Dissertation Award Committee
2022–	Co-founder and organizer of <i>Virtual Astronomy Software Talks</i> seminar series
2020–	Associate Editor for <i>Living Reviews in Computational Astrophysics</i>
2014–	OLCF User Group Executive Board (Elected to 3 year term 2014, re-elected in 2017; re-elected in 2020; re-elected in 2023; Vice chair: 2014–2015, 2018–2019; Chair: 2015–2016, 2019–2020)
ongoing	Referee for <i>Astronomy and Astrophysics</i> , the <i>Astrophysical Journal</i> , <i>Communications in Applied Mathematics and Computational Science</i> , <i>Computing in Science and Engineering</i> , <i>Journal of Computational Physics</i> , <i>Journal of Open Source Software</i> , <i>Journal of Open Source Education</i> , <i>Monthly Notices of the Royal Astronomical Society</i> , <i>Nature</i> , <i>Nuclear Physics A</i> , <i>Open Journal of Astrophysics</i> , <i>Open Research Europe</i> , and <i>Physical Review Letters</i>
2021	Served on a NASA Open Source Tools, Frameworks, and Libraries review panel
2020	External review committee member for Operational Assessment of the Oak Ridge Leadership Computing Facility (OLCF) (April 21–22, 2020)
2006–	Annual <i>Astronomy Open Night</i> public outreach talks, Stony Brook (Open Night coordinator from Fall 2013–Fall 2016)
2019	Reviewer for UK Science & Technology Facilities Council

2016–2019	Elected to the NERSC User’s Group Executive Committee (NUGEX)
2018	Reviewer for UK DiRAC HPC Facility
2018	Reviewer for Pazy Foundation / Israeli University Planning and Budgeting Committee and the Israeli Atomic Energy Commission (IAEC)
2016	Reviewer for Deutsche Forschungsgemeinschaft
2013, 2016, 2023	Served on a NASA ATP grant review panel
2011, 2014, 2016, 2018, 2020, 2022, 2023, 2024	External reviewer for DOE Office of Nuclear Physics
2014, 2016	External reviewer for NSF PRAC
2013	External reviewer for NSF Office of Cyber Infrastructure
2012	Reviewer for the Great Lakes Consortium for Petascale Computation (2012) proposals for the NCSA Blue Waters machine.
2007	External reviewer for NASA Astrophysics Theory and Fundamental Physics Program
2006	Served on NSF Astronomy and Astrophysics Program review panel

Meeting Organization

2025	Program Committee for the <i>Applications</i> track for the 40th IEEE International Parallel & Distributed Processing Symposium (IPDPS’26) meeting (New Orleans, May 2026)
2024	Program Committee for <i>Physics Domain</i> of the 2025 <i>Platform for Advanced Scientific Computing</i> (PASC) conference (Brugg, Switzerland, June 2025)
2023	Program Committee Co-Chair for <i>Physics Domain</i> of the 2024 <i>Platform for Advanced Scientific Computing</i> (PASC) conference (Zurich, Switzerland, June 2024)
2023	Co-chair of the Astronomy, Astrophysics, and Physics track of the SciPy 2023 meeting (July 2023)
2023	Program Committee Co-Chair for <i>Physics Domain</i> of the 2023 <i>Platform for Advanced Scientific Computing</i> (PASC) conference (Davos, Switzerland, June 2023)
2022	Organizer of the <i>Physics and Astrophysics of Common Envelopes</i> meeting (Los Alamos National Laboratory, May 2022)
2022	Co-chair of the Physics and Astronomy track of the SciPy 2022 meeting (July 2022)
2022	Program Committee for <i>Physics Domain</i> of the 2022 <i>Platform for Advanced Scientific Computing</i> (PASC) conference (Basel, Switzerland, June 2022)
2022	Organizer of APS April meeting DCOMP/DAP Invited Session <i>Frontiers in Computational Stellar Astrophysics</i> (NYC, April 2022)

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| 2021 | Co-chair of the Physics and Astronomy track of the SciPy 2021 meeting (virtual, July 2021) |
| 2021 | Co-organizer of a SIAM CSE 2021 mini-symposium <i>Performance Portability in Astrophysics Simulation Codes</i> (virtual, Feb. 2021) |
| 2020 | Co-chair of the Astronomy and Astrophysics track of the SciPy 2020 meeting (virtual, July 2020) |
| 2020 | Co-organizer of the <i>yAC: yt at CCA</i> meeting (Flatiron Institute / Center for Computational Astrophysics, March 2020) |
| 2019 | Scientific Organizing Committee, 2019 Compressible Convection Conference (Newcastle, UK, Sept. 2019) |
| 2018–2019 | Member of the SC19 Reproducibility Challenge track committee |
| 2017 | Co-organizer of the third <i>New York Area Computational Astrophysics meeting</i> (Flatiron Institute / Center for Computational Astrophysics, Sept. 2017) |
| 2016–2017 | Member of the Program Committee for the <i>13th International Workshop on OpenMP (IWOMP) 2017</i> (Stony Brook, NY 2017) |
| 2016 | Co-organizer of the second <i>New York Area Computational Astrophysics meeting</i> (American Museum of Natural History, April 2016) |
| 2015 | Scientific organizing committee for the workshop <i>GNASH: The anomalous metal-poor stars and convective-reactive nuclear astrophysics</i> (U. Victoria, Victoria, BC) |
| 2015 | Co-organizer of the <i>New York Area Computational Astrophysics meeting</i> (Farmingdale State College, April 2015) |
| 2014–2015 | Organizing committee for the 2015 <i>Oak Ridge Leadership Computing Facility User Meeting</i> |
| 2012–2013 | Local organizing committee for the <i>National Nuclear Physics Summer School</i> (NNPSS 2013). |
| 2012 | Co-convenor of <i>Thermonuclear explosions: Type Ias, Novae, and X-ray bursts</i> working group at <i>Nuclear Astrophysics Town Meeting</i> (Detroit, MI) |

Community Astrophysical Software / Other Projects

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| ongoing | Co-developer of the publicly-available low Mach number hydrodynamics code MAESTROeX, https://amrex-astro.github.io/MAESTROeX/ |
| ongoing | Co-developer of the publicly-available compressible (radiation-, magneto-) hydrodynamics code Castro, https://amrex-astro.github.io/Castro/ |
| ongoing | Creator and co-developer of the publicly-available teaching and prototyping hydrodynamics code pyro, https://github.com/python-hydro/pyro2/ |
| ongoing | Creator / co-developer of the pynucastro library, https://github.com/pynucastro/pynucastro |
| ongoing | Creator of the Open Astrophysics Bookshelf github organization http://open-astrophysics-bookshelf.github.io/ and author of the open text <i>Introduction to Computational Astrophysical Hydrodynamics</i> |

ongoing	Developed and distribute many simple teaching codes (advection, Eulerian compressible and incompressible hydro solvers, multigrid, etc., with accompanying notes and exercises), http://www.astro.sunysb.edu/mzingale/software/
ongoing	Created a library of astronomy animations introducing basic concepts (e.g. Kepler's laws, blackbody radiation, waves, binary star/exoplanet dynamics, etc.) as well as more advanced concepts (e.g. entropy in convection), http://zingale.github.io/astro_animations/ , also available on youtube, http://www.youtube.com/user/michaelzingale
ongoing	Contributor to and <i>project member</i> of the volumetric visualization package yt
2020–	Ombudsperson for the TARDIS Monte Carlo radiative transfer code (https://tardis-sn.github.io/tardis/team.html)
1997–2002	Original member of the FLASH Code development team

Guest/Visiting Appointments

2019–2020	Visiting Scholar at the Flatiron Institute / Center for Computational Astrophysics
2000–2003	Guest Appointment at Argonne National Laboratory / Mathematics and Computer Science Division
April 2001	Guest at the Max-Planck-Institut für Astrophysik

Professional Societies

Member of the American Astronomical Society

Member of the American Physical Society

Member of the Society for Applied and Industrial Mathematics

Students Advised

PhDs advised	Chris Malone (Stony Brook, PhD 2011, thesis: <i>Multidimensional Simulations of Convection Preceding a Type Ia X-ray Bursts</i>)
	Max Katz (Stony Brook, PhD 2016, thesis: <i>White Dwarf Mergers on Adaptive Meshes</i>)
	Adam Jacobs (Stony Brook, PhD 2016, thesis: <i>The Explosive Possibilities of Little Dwarfs: Low-Mach Number Modeling of Thin Helium Shells on Sub-Chandrasekhar Mass White Dwarfs</i>)
	Maria Guadalupe Barrios Sazo (Stony Brook, PhD 2020, thesis: <i>Studies toward the modeling of White Dwarf Mergers and Magnetohydrodynamics</i>)
	Xinlong Li (Stony Brook, PhD 2021, thesis: <i>3-d Simulation of Convection in an Electron-capture O-Ne Core</i>)
	Alexander Smith Clark (Stony Brook, PhD 2025, thesis: <i>Modeling Astrophysical Reactions with Applications in Multidimensional Classical CO-Novae Simulations</i>)

	Eric Johnson (Stony Brook, PhD 2025, thesis: <i>Developing Multidimensional Simulations of H/He Flames in Type I X-ray Bursts</i>)
Masters students advised	Mu-Hung Chang (Stony Brook, MA 2017, thesis: <i>Application of Spectral Deferred Correction for 1-D Astrophysical Detonation</i>) Hengrui Zhan (Stony Brook, MA 2019, thesis: <i>Implementation of an Improved Multipole Expansion Method</i>) Zhi Chen (Stony Brook, MA 2023, thesis: <i>Sensitivity of He Flames in X-ray Bursts to Nuclear Physics</i>) Sam Glosser (Stony Brook, MA 2024, thesis: <i>Identifying the Classical Thermonuclear Regime of Plasma Screening in Astrophysical Flows</i>)
postdocs advised	Alice Harpole (worked on Maestro rotation support, GPU acceleration, algorithm development, massive star evolution).
current grad students	Khanak Bhargava Zhi Chen Melissa Rasmussen
undergrad honors theses	Luke Nolan (Stony Brook, BS 05/2016, thesis: <i>Flame Wave Propagation on the Surface of Neutron Stars During Type I X-Ray Bursts</i>) Abigail Bishop (Stony Brook, BS 05/2019, thesis: <i>Expanding the Modeling of Type Ia Supernovae</i>) Kiran Eiden (Stony Brook, BS 05/2020, thesis: <i>Propagation of Thermonuclear Flame Fronts in Type I X-ray Bursts</i>) Sarah Cocuzza (Stony Brook, BS 05/2025, thesis: <i>Simulating Convection with Pyro Code, Is a More Thorough Solver Worth the Wait?</i>) Serena Kinney (Stony Brook, BS 05/2026, thesis: <i>Alternative Models for H/He and He Flames in X-Ray Bursts</i>) Carolyn Reilly (Stony Brook, BS 05/2026, thesis: <i>Production of Nickel in Type Ia Supernovae</i>)

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Michael Zingale / Publications and Talks

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79. *Multi-dimensional Models of Mixed H/He Flames in X-ray Bursts*, E. T. Johnson & M. Zingale, 2025, submitted to Open Journal of Astrophysics
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77. *Approximating Convective Urca Cooling in a Simmering White Dwarf*, B. Boyd, A. C. Calder, D. Townsley, & M. Zingale, 2025, Journal of Physics: Conference Series, 2997, 1, 012006
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A. C. Calder, B. C. Curtis, L. J. Dursi, B. Fryxell, G. Henry, P. MacNeice, K. Olson, P. Ricker, R. Rosner, F. X. Timmes, H. M. Tufo, J. W. Truran, & M. Zingale, 2000, Gordon Bell Prize winner/Special category, SC '00: Proceedings of the 2000 ACM/IEEE Conference on Supercomputing
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3. *On the Cellular Structure of Carbon Detonations*,
F. X. Timmes, M. Zingale, K. Olson, B. Fryxell, P. Ricker, A. C. Calder, L. J. Dursi, J. W. Truran, & R. Rosner, 2000, ApJ, 543, 938
DOI: 10.1086/317135
2. *FLASH: An Adaptive Mesh Hydrodynamics Code for Modeling Astrophysical Thermonuclear Flashes*,

B. Fryxell, K. Olson, P. Ricker, F. X. Timmes, M. Zingale, D. Q. Lamb, P. MacNeice, R. Rosner, & H. Tufo, 2000, *ApJS*, 131, 273
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1. *Flash Code: Studying Astrophysical Thermonuclear Flashes,*

R. Rosner, A. Calder, J. Dursi, B. Fryxell, D. Q. Lamb, J. C. Niemeyer, K. Olson, P. Ricker, F. X. Timmes, J. Truran, H. Tufo, Y. Young, M. Zingale, E. Lusk, & R. Stevens, 2000, *CiSE*, 2, 33
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28. *Numerical Treatment of Shock-induced Nuclear Burning in Double Detonation Type Ia Supernovae,*
R. Brady & M. Zingale, 2025, *Research Notes of the AAS*, 9, 5, 113
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27. *Well-Balanced Hydrodynamics for the Piecewise Parabolic Method with Characteristic Tracing,*
M. Zingale, 2024, *Research Notes of the AAS*, 8, 9, 219
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26. *A Fully Explicit Integrator for Modeling Astrophysical Reactive Flows,*
P. Johnson, M. Zingale, E. T. Johnson, A. Smith, & K. Niemeyer, 2023, *Research Notes of the AAS*, 7, 12, 282
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25. *Practical Effects of Integrating Temperature with Strang Split Reactions,*
M. Zingale, M. P. Katz, D. E. Willcox, & A. Harpole, 2021, *Research Notes of the AAS*, 5, 4, 71
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24. *The LOFT mission concept: a status update,*
M. Feroci et al., 2016, *Proc. SPIE 9905, Space Telescopes and Instrumentation 2016: Ultraviolet to Gamma Ray*, 99051R, July 25, 2016
DOI: 10.48550/arXiv.1501.02776
23. *Understanding Ignition in Type Ia Supernovae,*
M. Zingale, A. Jacobs, A. S. Almgren, J. B. Bell, A. Nonaka, C. Malone, & S. Woosley, 2015, extended abstract for the 25th International Colloquium on the Dynamics of Explosions and Reactive Systems, Leeds, UK, Aug. 2–7, 2015
22. *Low Mach Number Modeling of Stratified Flows,*
A. S. Almgren, J. B. Bell, A. Nonaka, & M. Zingale, 2014, in *Finite Volumes for Complex Applications VII: Methods, Theoretical Aspects—FVCA 7*, Berlin, June 2014, ed. Fuhrmann, J., Ohlberger, M., & Rohde, C., 3–15; *Proceedings of the FVCA7 - The International Symposium of Finite Volumes for Complex Applications VII Berlin*, June 15–20, 2014
21. *From Convection to Explosion: End-to-End Simulation of Type Ia Supernovae,*
A. Nonaka, A. S. Almgren, J. B. Bell, H. Ma, S. E. Woosley, & M. Zingale, 2011, *Proceedings of SciDAC 2011*, Denver, CO, July 10–14, 2011

20. *MAESTRO, CASTRO, and SEDONA — Petascale Codes for Astrophysical Applications*,
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& M. Zingale, 2010, Proceedings of the 2010 Scientific Discovery through Advanced Comput-
ing (SciDAC) Conference. Chattanooga, Tennessee, July 11–15, 2010. Oak Ridge National
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19. *Type Ia Supernovae: Advances in Large Scale Simulation*,
H. Ma, M. Zingale, S. E. Woosley, A. J. Aspden, J. B. Bell, A. S. Almgren, A. Nonaka, & S. Dong,
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17. *Astrophysical Applications of the Maestro Code*,
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F. Röpke, V. Sankaran, & M. Zingale, 2007, Proceedings of SciDAC 2007, Journal of Physics:
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13. *Efficiency Gains from Time Refinement on AMR Meshes and Explicit Timestepping*,
L. J. Dursi & M. Zingale, 2005, Adaptive Mesh Refinement—Theory and Applications, Pro-
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11. *Simulations of Rising Hydrodynamic and Magnetohydrodynamic Bubbles*,
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10. *Investigations of Pointwise Ignition of Helium Deflagrations on Neutron Stars*,
M. Zingale, S. E. Woosley, A. Cumming, A. Calder, L. J. Dursi, B. Fryxell, K. Olson, P. Ricker, R. Rosner, & F. X. Timmes, 2002, 3D Stellar Evolution, ASP Conference Proceedings, Vol. 293, 22–26 July 2002 at UC Davis, Livermore, CA, Ed. by S. Turcotte, S. C. Keller, & R. M. Cavallo.
9. *Onset of Convection on a Pre-Runaway White Dwarf*,
L. J. Dursi, A. C. Calder, A. Alexakis, J. W. Truran, M. Zingale, B. Fryxell, P. Ricker, F. X. Timmes, & K. Olson, 2002, Classical Nova Explosions: International Conference on Classical Nova Explosions. AIP Conference Proceedings, Vol. 637. Sitges, Spain, 20–24 May, 2002. Edited by M. Hernanz & J. Jose
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8. *Mixing by Non-linear Gravity Wave Breaking on a White Dwarf Surface*,
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7. *Mixing by Wave Breaking at the Surface of a White Dwarf*,
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5. *Quenching Processes in Flame-Vortex Interactions*,
M. Zingale, J. C. Niemeyer, F. X. Timmes, L. J. Dursi, A. C. Calder, B. Fryxell, D. Q. Lamb, K. Olson, P. Ricker, R. Rosner, J. W. Truran, & P. MacNeice, 2001, 20th Texas Symposium on Relativistic Astrophysics, Austin, Texas, 10–15 Dec. 2000, Melville, NY: AIP Conference Proceedings, Vol. 586. Edited by J. C. Wheeler & H. Martel, also AIP Conference Series 586, 490–492.
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4. *Simulations of Astrophysical Fluid Instabilities*,
A. C. Calder, B. Fryxell, R. Rosner, L. J. Dursi, K. Olson, P. M. Ricker, F. X. Timmes, M. Zingale, P. MacNeice, & H. M. Tufo, 2001, 20th Texas Symposium on Relativistic Astrophysics, Austin,

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3. *Adaptive Mesh Simulations Of Astrophysical Detonations Using the ASCI Flash Code,*

B. Fryxell, A. C. Calder, L. J. Dursi, D. Q. Lamb, P. MacNeice, K. Olson, P. M. Ricker, R. Rosner, F. X. Timmes, J. W. Truran, H. M. Tufo, & M. Zingale, Proceedings of the VII International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT 2000), Fermilab, October 16–20, 2000.

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2. *Large-Scale Simulations of Clusters of Galaxies,*

P. M. Ricker, A. C. Calder, L. J. Dursi, B. Fryxell, D. Q. Lamb, P. MacNeice, K. Olson, R. Rosner, F. X. Timmes, J. W. Truran, H. M. Tufo, & M. Zingale, Proceedings of the VII International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT 2000), Fermilab, October 16–20, 2000.

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1. *Helium Detonations on Neutron Stars,*

B. Fryxell, M. Zingale, F. X. Timmes, D. Q. Lamb, K. Olson, A. C. Calder, L. J. Dursi, P. Ricker, R. Rosner, J. W. Truran, P. MacNeice, & H. Tufo, Proceedings of the 10th Workshop on “Nuclear Astrophysics”, Ringberg Castle, Tegernsee, Germany, March 20–25 2000.

Open Books

2. *Teaching and Learning with Jupyter,*

L. A. Barba, L. J. Barker, D. S. Blank, J. Brown, A. B. Downey, T. George, L. J. Heagy, K. T. Mandli, J. K. Moore, D. Lippert, K. E. Niemeyer, R. R. Watkins, R. H. West, E. Wickes, C. Willing, & M. Zingale

<https://jupyter4edu.github.io/jupyter-edu-book/>

1. *Introduction to Computational Astrophysical Hydrodynamics,*

M. Zingale

https://github.com/Open-Astrophysics-Bookshelf/numerical_exercises

White Papers

8. *Multidisciplinary Science in the Multimessenger Era,*

E. Burns et al., 2025, White Paper of the Third Time-Domain And Multimessenger Astrophysics workshop: Multidisciplinary Science in the Multimessenger Era

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7. *Horizons: Nuclear Astrophysics in the 2020s and Beyond,*

H. Schatz et al., 2022, Journal of Physics G Nuclear Physics, 49, 11, 110502

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6. *MMA SAG: Thermonuclear Supernovae*,
M. Zingale, C. Fryer, A. Hungerford, S. Safi-Harb, R. Trappitsch, R. Fisher, A. Calder, & K. Shen, *Astro2020: Decadal Survey on Astronomy and Astrophysics*, science white papers, no. 259; *Bulletin of the American Astronomical Society*, Vol. 51, Issue 3, id. 259 (2019)
5. *Nuclear Physics Exascale Requirements Review: An Office of Science review sponsored jointly by Advanced Scientific Computing Research and Nuclear Physics*,
J. Carlson et al. 2017, US Department of Energy, Washington, DC (United States). *Advanced Scientific Computing Research and Nuclear Physics*
4. *The Importance of Computation in Astronomy Education*,
M. Zingale, F. X. Timmes, R. Fisher, & B. W. O'Shea, 2016, white paper submitted to the AAS Education Taskforce call
(<https://aas.org/posts/opportunity/2016/04/aas-task-force-education-begins-its-work>)
3. *White Paper on Nuclear Astrophysics*,
A. Arcones et al. 2016, Community white paper based on 2012 JINA Town Meeting in Detroit, MI, and 2014 APS Town Meeting in College Station, TX
2. *Modeling Astrophysical Explosions with Sustained Exascale Computing*,
M. Zingale, A. C. Calder, C. M. Malone, & F. X. Timmes, 2015, Response to RFI NOT-GM-15-122: *Science Drivers Requiring Capable Exascale High Performance Computing*
1. *The LOFT perspective on neutron star thermonuclear bursts*,
J. J. M. in 't Zand, D. Altamirano, D. R. Ballantyne, S. Bhattacharyya, E. F. Brown, Y. Cavocchi, D. Chakrabarty, J. Chenevez, A. Cumming, N. Degenaar, M. Falanga, D. K. Galloway, A. Heger, J. José, L. Keek, M. Méndez, S. Mahmoodifar, M. Linares, C. M. Malone, M. C. Miller, F. B. S. Paerels, J. Poutanen, A. Różańska, H. Schatz, M. Serino, V. F. Suleimanov, T. E. Strohmayer, F.-K. Thielemann, A. L. Watts, N. N. Weinberg, S. E. Woosley, W. Yu, S. Zhang, & M. Zingale, 2015, White Paper in Support of the Mission Concept of the Large Observatory For x-ray Timing

Invited Lectures / Seminars / Colloquia

- | | |
|------------|---|
| 06/10/2026 | Invited Lecturer for the MESA Summer School 2026 (Jackson, WY) <i>All-in On Urca</i> |
| 05/22/2026 | Invited talk at the 2026 Frontier in Nuclear Astrophysics Meeting (Texas A&M), <i>Building a community python library for nuclear astrophysics</i> |
| 03/19/2026 | Invited talk in the AMReX session for HPSFCon, Chicago, IL, <i>Modeling Astrophysical Reactive Flows with AMReX-Astrophysics Castro</i> |
| 02/12/2026 | Seminar at AMNH, NY, NY, <i>What I Learned About Supercomputing from Blowing Up Stars</i> |
| 05/06/2025 | Invited Keynote at Cray Users Group 2025 meeting, Jersey City, NJ, <i>What I Learned About Supercomputing from Blowing Up Stars</i> |
| 10/09/2024 | Invited speaker at APS-DNP 2024 meeting, Mini-Symposium: Nuclear Data in the Cosmos, <i>pynucastro: A python library for connecting nuclear data to astrophysical simulations</i> |

- 09/13/2024 IReNA Online Seminar Series, *pynucastro: A python library connecting nuclear experiment data and astrophysics*
- 08/07/2024 Invited talk at the Rise_Time conference, Purdue University, *Modeling Astrophysical Reacting Flows*
- 07/30/2024 Modeling Talk seminar series at Google research / x.company, *The Challenges of Modeling Stellar Explosions*
- 07/04/2024 Invited talk at AstroNum 2024, La Rochelle, France, *Strong Coupling of Hydrodynamics and Reactions in Astrophysical Flows*
- 04/15/2024 Colloquium at Temple University, *Modeling Stellar Explosions*
- 03/06/2024 Invited minisymposium talk in the Performance Optimization for Multiphysics Applications on Structured Mesh with Particles minisymposium at the 2024 SIAM Conference on Parallel Processing for Scientific Computing, *Modeling Astrophysical Reactive Flows with Castro at the Exascale*
- 11/04/2023 Invited talk at APS-MAS, University of Delaware, *Engaging Students in Computational (Astro)Physics Courses and Research*
- 10/23/2023 Invited talk at the Interfaces and Mixing, Exploration Conference, 23-26 Oct 2023, Kavli Institute for Theoretical Physics, *Convective Mixing in Stellar Reactive Flows*
- 06/28/2023 Invited talk at AstroNum 2023–15th International Conference on Numerical Modeling of Space Plasma Flows, Pasadena, CA, USA, *Coupling Hydrodynamics and Astrophysical Reaction Networks*
- 11/21/2022 Invited talk in *Astrophysical Turbulence: Current Understanding and Modeling Challenges* minisymposium at the APS/Division of Fluid Dynamics 2022 meeting, *Modeling Astrophysics Reactions in Turbulent Environments*
- 11/14/2022 Invited talk a HPC session of the Computing Working Group at the Argonne Town Meeting on Nuclear Structure, Reactions, and Astrophysics, *Blowing Up Stars Over the Next Decade*
- 07/28/2022 Universiti Malaya (Kuala Lumpur, Malaysia), Department of Physics Colloquium, *Modeling Astrophysical Thermonuclear Explosions*
- 05/20/2022 FastMath seminar, *Algorithmic Improvements for Coupling Hydrodynamics and Reactions in Astrophysical Flows*
- 05/04/2022 Invited speaker in the AMReX Breakout Session of the 2022 ECP Annual Meeting, *Castro Developments for Exascale Platforms*
- 03/16/2022 Invited Focus Session Speaker in *Extreme-Scale Computational Science Discovery in Fluid Dynamics and Related Disciplines*, APS March Meeting, *The Challenges of Modeling Astrophysical Reactive Flows*
- 11/17/2021 Invited panelist for KITP Transport in Stars workshop on convection (KITP, Santa Barbara, CA)
- 10/19/2021 Astronomy Society of Ireland Colloquium, *The Challenges of Modeling Astrophysical Reactive Flows*
- 10/18/2021 Michigan State University, Department of Computational Mathematics, Science and Engineering Colloquium, *The Challenges of Modeling Astrophysical Reactive Flows*

- 03/02/2021 Invited talk in the *Computational Methods in Explosive Nuclear Astrophysics* session at the SIAM Computational Science and Engineering 2021 meeting, *Modeling Astrophysical Reacting Flows*
- 01/30/2020 Princeton University, Department of Astrophysical Sciences Thunch talk, *Modeling Stellar Explosions*
- 01/09/2020 Invited presentation at Texas Advanced Computing Center *Workshop on Future Directions in Extreme Scale Computing for Scientific Grand Challenges on Challenges in Modeling Astrophysical Thermonuclear Explosions*
- 11/06/2019 Stony Brook University, Department of Physics and Astronomy Colloquium, *Modeling Thermonuclear X-ray Bursts*
- 07/05/2019 Invited talk at AstroNum 2019—14th International Conference on Numerical Modeling of Space Plasma Flows, Paris, France, *Improved Coupling of Hydrodynamics and Nuclear Burning in Astrophysical Flows using SDC*
- 02/26/2019 Invited talk in the *Spectral Deferred Correction Methods for Temporal Integration* session at the SIAM Computational Science and Engineering 2019 meeting, *Improved Coupling of Hydrodynamics and Nuclear Burning in Astrophysical Flows using SDC*
- 10/12/2018 Flatiron Institute Center for Computational Astrophysics Colloquium, *Algorithmic Demands for Modeling X-ray Bursts and Type Ia Supernovae*
- 08/23/2018 Talk at the TEAMS Collaboration meeting, *StarKiller Microphysics*
- 06/26/2018 Invited talk at AstroNum 2018—13th International Conference on Numerical Modeling of Space Plasma Flows, Panama City, Florida, *Modeling X-ray Bursts with the AMReX Astrophysics Suite*
- 08/10/2017 Seminar at LLNL High Energy Density Science Center, LLNL, *Modeling Stellar Explosions with the AMReX Astrophysics Suite*
- 07/27/2017 Seminar at Computational Science Initiative, BNL, *The AMReX Astrophysics Suite: Simulating the Stars at the Exascale*
- 06/30/2017 Invited talk at AstroNum 2017—12th International Conference on Numerical Modeling of Space Plasma Flows, St. Malo, France, *Computational Challenges of Modeling X-ray Bursts and Type Ia Supernovae*
- 06/02/2017 Invited participant / overview talk at Stellar Hydro Days, University of Victoria, *Modeling Stellar Convection and Explosions with Maestro, Castro, and the BoxLib/AMReX Astrophysics Suite*
- 04/05/2017 Astronomy Seminar at Michigan State University, *Computational Challenges of Modeling X-ray Bursts and Type Ia Supernovae*
- 02/23/2017 Seminar at Stony Brook Institute for Advanced Computational Science, *Computational Challenges of Modeling X-ray Bursts and Type Ia Supernovae*
- 06/15/2016 Case study talk at DOE Nuclear Physics / ASCR Exascale Requirements Review, Gaithersburg, MD, *Thermonuclear Transients*
- 04/29/2016 Seminar at Oak Ridge National Laboratory, *Modeling Stellar Explosions with Maestro, Castro, and the BoxLib Astrophysics Suite*

- 03/17/2016 Talk at the *18th Workshop on Nuclear Astrophysics*, Ringberg Castle, Tegernsee, Germany, *Models of convection in X-ray bursts and pre-SNe Ia white dwarfs*
- 02/26/2016 Seminar at the U. S. Naval Research Laboratory, *Computational Challenges of Modeling X-ray Bursts and Type Ia Supernovae*
- 08/02/2015 Invited talk at the *International Colloquium on the Dynamics of Explosions and Reactive Systems (ICDERS)*, Leeds, UK, *Understanding Ignition in Type Ia Supernovae*
- 06/22/2015 Invited talk at the *OLCF User's Meeting*, ORNL, Oak Ridge, TN, *Computation Challenges of Modeling Astrophysical Explosions*
- 06/03/2015 Invited talk at the *Fifty One Ergs* meeting, NCSU, *Modeling the Early Phases of Type Ia Supernovae*
- 05/24/2015 "Setting the Stage" talk on *Stellar Hydrodynamics* at the JINA GNASH: *The anomalous metal-poor stars and convective-reactive nuclear astrophysics workshop*, Victoria, BC, Canada, <http://jina-cee.phys.uvic.ca/gnash-workshop/talks-and-contributions/monday/setting-the-stage>
- 04/08/2015 Seminar at U Mass Dartmouth, *Algorithmic Developments for Modeling Stellar Explosions*
- 01/15/2015 CCS-2 Seminar at Los Alamos National Laboratory, *The Challenges of Modeling Type Ia Supernovae and X-ray Bursts*
- 09/15/2014 Invited talk at the *Type Ia Supernovae: progenitors, explosions, and cosmology conference*, Chicago, IL, *Modeling the Early Phases of SNe Ia*, <https://kicp-workshops.uchicago.edu/sn2014/presentations.php>
- 04/30/2014 Invited presentation at *Large Scale Computing and Storage Requirements for Nuclear Physics (NP): Target 2017 meeting*, *Convection in X-ray Bursts*
- 02/28/2014 Astronomy Seminar at the Center for Cosmology and Particle Physics, New York University, *Modeling Convective Burning in Type Ia Supernovae and X-ray Bursts*
- 09/27/2013 Nuclear Theory Seminar at Brookhaven National Lab, *Modeling Convective Burning in Type Ia Supernovae and X-ray Bursts*
- 07/09/2013 Seminar at the Flash Center, University of Chicago, *Modeling Convective Burning in Type Ia Supernovae and X-ray Bursts*
- 10/10/2012 Astro Computation working group at 2012 *Nuclear Astrophysics Town Meeting*, *Thermonuclear Driven Events*
- 04/04/2012 Nuclear Astrophysics Seminar at Ohio University entitled *The Challenges of Modeling Explosive Phenomena*
- 07/28/2010 Invited talk at the Lorentz Center Workshop on *X-ray Bursts and Burst Oscillations* entitled *The Algorithmic Challenges of Multidimensional Models of X-ray Bursts*, <http://www.lorentzcenter.nl/lc/web/2010/408/info.php3?wsid=408>
- 05/13/2010 Joint NRAO / UVa Dept. of Astronomy Colloquium (Charlottesville, VA) entitled *Modeling Convection and Ignition in Type Ia Supernovae*
- 03/31/2010 Center for the Study of Cosmic Evolution Seminar, Dept. of Physics and Astronomy, Michigan State University (E. Lansing, MI), entitled: *Modeling Convection and Ignition in Type Ia Supernovae*

- 05/12/2009 Astronomy Seminar at the American Museum of Natural History (New York, NY), entitled: *Modeling Convection and Ignition in Type Ia Supernovae*
- 09/30/2008 Astronomy Seminar at the Institute for Advanced Studies (Princeton, NJ), entitled: *New Methods for Modeling Type Ia Supernovae*
- 07/15/2008 Invited Poster at the SciDAC 2008 conference (Seattle, WA), entitled: *Astrophysical Applications of the Maestro Code* (with co-authors: A. S. Almgren, J. B. Bell, C. M. Malone, & A. J. Nonaka)
- 04/06/2007 Astronomy Seminar at Rutgers University (New Brunswick, NJ), entitled: *The Challenges of Modeling Type Ia Supernova*
- 10/31/2006 Astronomy Colloquia at McGill University (Montreal, CA), entitled: *Understanding Type Ia Supernovae*
- 06/27/2006 Invited talk at the SciDAC 2006 conference (Denver, CO), entitled: *The Challenges of Modeling Type Ia Supernovae*
- 10/03/2005 T-13 Seminar, Los Alamos National Laboratory, entitled: *Simulations of Thermonuclear Flames in Type Ia Supernovae*
- 06/26/2005 Invited poster at the SciDAC 2005 conference (San Francisco, CA), *The Physics of Thermonuclear Flames in Type Ia Supernovae*
- 03/01/2005 Astronomy Seminar at SUNY Stony Brook, *Flame Instabilities in Type Ia Supernovae*
- 02/23/2005 N Division Seminar, Lawrence Livermore National Laboratory, *Flame Instabilities in Type Ia Supernovae*
- 12/17/2003 Astrophysics Seminar, Institute for Advanced Study, Princeton, NJ, *Flame Instabilities in Type Ia Supernovae*

Popular Press Features

How Stars Explode, Forbes.com, Oct. 1, 2009

(<http://www.forbes.com/2009/09/30/supernovae-universe-science-technology-breakthroughs-stars.html>)

Unveiled: The First Full 3-D Model of a Star Going Supernova, Popular Science Online, Sept. 24, 2009

(<http://www.popsci.com/military-aviation-amp-space/article/2009-09/first-3-d-models-white-dwarf-supernova>)

Flash Upon a Neutron Star, American Scientist, Sept.–Oct. 2000, vol. 88, no. 5, p. 400.

Popular Press Mentions

Stars Go Kaboom, Spilling Cosmic Secrets, Science News, 2009, Vol. 176, #4 (Aug. 15, 2009)

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